Ajay Venkatesh

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Education

New York University, New York, NY — *Aug 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

- **Campus Mesh** — *Jan 2026 – Present*
  Software Engineer — New York, US
  - Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management, backed by a **MongoDB** store.
  - Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing engagement by **28%** and retention by **50%**.
  - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented **system design** for approval and established Role-Based Access **(4+ roles)** via **OAuth, Midway SSO, HELIS**.
  - Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM; reduced provisioning time **70%** with **multi-stage, multi-region CI/CD**.
  - Built a **React 18** frontend with optimized API integration, advanced filtering, and client-side caching, reducing page load time by **40%**.
  - Engineered the backend services in **Java** (Coral, Smithy, Dagger, CBOR); applied **low-level design patterns**, cutting backend execution time by **35%** with unit/integration testing via **Mockito**.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  Software Developer — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
  - Developed an insurance analytics platform (**React, Node.js, REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue across **6 clients**.
  - Streamlined underwriting via a broker portal integrated with insurer platforms; implemented **document/media processing** and **WebSocket-based real-time communication**.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Developed a distributed backend on **Microsoft Azure** for a health platform integrating with **Microsoft CRM**, with event-driven processing and fault-tolerant retries over **SQL Server** stores.
  - Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
  - Optimized data ingestion with **parallel processing** on **100K-row** Excel feeds, cutting execution from **5 hours to 1 hour**.
  - Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.

Projects

- **Advanced Expense Tracker (Python Desktop Application)** | *Python, Tkinter, SQLite3, Pandas, Matplotlib*
  - Architected a modular **Python application** for individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns** over **SQLite3**.
  - Implemented concurrent transaction handling and real-time cost splitting using **Python threading** and locks for thread-safe DB writes; visualized spending trends with **Matplotlib**.
- **Hospital Management System (Full-Stack Application)** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
  - Built a **full-stack healthcare platform** (React, Node.js, Express, MongoDB) managing appointments, patient records, lab reports, and medical history with **role-based access control**.
  - Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction and deployed on **AWS SageMaker** for real-time diagnostic support.
- **InviLite** | *AWS, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
  - Built a **RAG system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%**.

Skills

- **Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices, HTML, CSS, Sequelize
- **Databases:** MySQL, PostgreSQL, DynamoDB, MongoDB, FireBase, Dataverse
- **Cloud & DevOps:** AWS (CDK, EC2, S3, DynamoDB, API Gateway, Lambda, Cognito, IAM, SageMaker), Azure (OpenAI, Function Apps, Storage), Docker, Kubernetes, CI/CD
- **AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, Reinforcement Learning, Hugging Face
- **Tools & Platforms:** Git, Linux, Visual Studio, Cursor, GitHub Copilot, Jupyter, Postman, Power BI, Training optimization, Model robustness, Explainability, Compiler design, Innovation

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals